Step 1 — Environment Setup

```
# @title Step 1 — Environment Setup
!pip install transformers
!pip install datasets
!pip install torch
!pip install scikit-learn
```

```
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from datasets) (3.24.3)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.0.2)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from datasets) (2.2.2)
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.32.4)
Requirement already satisfied: tqdm>=4.66.3 in /usr/local/lib/python3.12/dist-packages (from datasets) (4.67.3)
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets) (3.6.0)
Requirement already satisfied: multiprocess<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=
Requirement already satisfied: huggingface-hub>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (1.5.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from datasets) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from datasets) (6.0.3)
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24.0->
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24.0->datasets) (
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2-
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datas
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datas
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.3)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub>=0
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingfac
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0-
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->d
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>=0.24.
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>=0.24.
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hu
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer-
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->type
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.24.3)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: numpy>=1.19.5 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (2.0.2)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
```

Import libraries

```
# @title Import libraries
import torch
from transformers import AutoTokenizer, AutoModelForSequenceClassification
from datasets import load_dataset
```

Step 2 — Data Loading

```
# @title Step 2 — Data Loading
dataset = load_dataset("imdb")
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
README.md: 7.81k/? [00:00<00:00, 415kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
plain_text/train-00000-of-00001.parquet: 100% 21.0M/21.0M [00:01<00:00, 20.7MB/s]
plain_text/test-00000-of-00001.parquet: 100% 20.5M/20.5M [00:00<00:00, 178MB/s]
plain_text/unsupervised-00000-of-00001.p(…): 100% 42.0M/42.0M [00:00<00:00, 129MB/s]
Generating train split: 100% 25000/25000 [00:00<00:00, 94660.57 examples/s]
Generating test split: 100% 25000/25000 [00:00<00:00, 84268.94 examples/s]
Generating unsupervised split: 100% 50000/50000 [00:00<00:00, 119318.56 examples/s]
```

Step 3 — Text Preprocessing

```
# @title Step 3 — Text Preprocessing
def preprocess(example):
    return tokenizer(example['text'], truncation=True, padding='max_length')
```

Step 4 — Tokenization

```
# @title Step 4 — Tokenization
tokenizer = AutoTokenizer.from_pretrained("distilbert-base-uncased")
```

```
config.json: 100% 483/483 [00:00<00:00, 10.6kB/s]
tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 3.08kB/s]
vocab.txt: 232k/? [00:00<00:00, 3.19MB/s]
tokenizer.json: 466k/? [00:00<00:00, 4.06MB/s]
```

Step 5 — Train Test Split

```
# @title Step 5 — Train Test Split
train_dataset = dataset["train"]
test_dataset = dataset["test"]
```

Step 6 — Create PyTorch Dataset Class

```
# @title Step 6 — Create PyTorch Dataset Class
tokenized_train_dataset = train_dataset.map(preprocess, batched=True)
tokenized_test_dataset = test_dataset.map(preprocess, batched=True)

tokenized_train_dataset.set_format(type='torch', columns=['input_ids','attention_mask','label'])
tokenized_test_dataset.set_format(type='torch', columns=['input_ids','attention_mask','label'])
```

```
Map: 100% 25000/25000 [00:31<00:00, 733.84 examples/s]
Map: 100% 25000/25000 [00:29<00:00, 852.74 examples/s]
```

Step 7 — Build the Model

```
# @title Step 7 — Build the Model
model = AutoModelForSequenceClassification.from_pretrained(
    "distilbert-base-uncased",
    num_labels=2
)
```

```
model.safetensors: 100%                                    268M/268M [00:02<00:00, 167MB/s]

Loading weights: 100%                    100/100 [00:00<00:00, 360.53it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.weight]
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                     | Status     |
------------------------+------------+-
vocab_layer_norm.weight | UNEXPECTED |
vocab_projector.bias    | UNEXPECTED |
vocab_transform.bias    | UNEXPECTED |
vocab_layer_norm.bias   | UNEXPECTED |
vocab_transform.weight  | UNEXPECTED |
classifier.bias         | MISSING    |
pre_classifier.bias     | MISSING    |
pre_classifier.weight   | MISSING    |
classifier.weight       | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING   :those params were newly initialized because missing from the checkpoint. Consider training on your downstream t
```

Step 8 — Model Training

```
# @title Step 8 — Model Training
from transformers import Trainer, TrainingArguments

training_args = TrainingArguments(
    output_dir="./results",
    num_train_epochs=3,
    per_device_train_batch_size=8,
)

trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=tokenized_train_dataset
)

trainer.train()
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as tru
  super().__init__(loader)
```

[286/9375 1:59:07 < 63:32:35, 0.04 it/s, Epoch 0.09/3]

Step	Training Loss

Step 9 — Model Evaluation

```
# @title Step 9 — Model Evaluation
trainer.evaluate()
```